

Solve $a_n = 4a_{n-1}$, $n \geq 1$, $a_0 = 2$ by Generating function

Soln

Let $G(x)$ be G.F. for $\{a_n\}$, $n \geq 0$

$$G(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$$

Multiply given eqn with x^n

$$\begin{aligned} a_n x^n &= 4 a_{n-1} x^n \\ &= 4x a_{n-1} x^{n-1} \end{aligned}$$

$$\sum_{n=1}^{\infty} a_n x^n = 4x \sum_{n=1}^{\infty} a_{n-1} x^{n-1}$$

$$a_0 + \sum_{n=1}^{\infty} a_n x^n = a_0 + 4x \sum_{n=1}^{\infty} a_{n-1} x^{n-1}$$

$$\sum_{n=0}^{\infty} a_n x^n = 2 + 4x \sum_{n=1}^{\infty} a_{n-1} x^{n-1}$$

$$\Rightarrow G(x) = 2 + 4x G(x)$$

$$\Rightarrow G(x) [1 - 4x] = 2$$

$$\Rightarrow G(x) = \frac{2}{1-4x} = 2(1-4x)^{-1}$$

$$= 2[1 + 4x + 4^2 x^2 + \dots + 4^n x^n + \dots]$$

$$\Rightarrow a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n + \dots =$$

$$2 + 2 \cdot 4x + 2 \cdot 4^2 x^2 + \dots$$

$$+ 2 \cdot 4^n x^n + \dots$$

$$\therefore \boxed{a_n = 2 \cdot 4^n}, \quad n \geq 0$$

Ex

$$a_n - 3a_{n-1} = 2$$

$$\forall n \geq 1, a_0 = 2$$

Soln

$$\text{Let } G(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$a_n - 3a_{n-1} = 2$$

$$a_n x^n - 3a_{n-1} x^n = 2x^n$$

$$\Rightarrow \sum_{n=1}^{\infty} a_n x^n - 3x \sum_{n=1}^{\infty} a_{n-1} x^{n-1} = 2 \sum_{n=1}^{\infty} x^n$$

$$\Rightarrow a_0 + \sum_{n=1}^{\infty} a_n x^n - 3x \sum_{n=1}^{\infty} a_{n-1} x^{n-1} = a_0 + 2(x + x^2 + x^3 + \dots)$$

$$\Rightarrow \sum_{n=0}^{\infty} a_n x^n - 3x G(x) = 2 + \frac{2x}{1-x}$$

$$\Rightarrow G(x)(1-3x) = \frac{2}{1-x}$$

$$G(x) = \frac{2}{(1-3x)(1-x)} = \frac{-1}{1-x} + \frac{3}{1-3x}$$

$$= -(1-x)^{-1} + 3(1-3x)^{-1}$$

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$$= -(1+x+x^2+\dots+x^n+\dots) + 3(1+3x+3^2x^2+\dots+3^nx^n+\dots)$$

$$a_n = -1 + 3 \cdot 3^n, \quad n \geq 0$$

$$= -1 + 3^{n+1}, \quad n \geq 0$$

Solve $a_{n+2} - 2a_{n+1} + a_n = 2^n$, $a_0 = 2$, $a_1 = 1$

Let $G(x) = \sum_{n=0}^{\infty} a_n x^n$ be the generating function

$$a_{n+2} x^n - 2a_{n+1} x^n + a_n x^n = 2^n x^n$$

$$\sum_{n=0}^{\infty} a_{n+2} x^n - 2 \sum_{n=0}^{\infty} a_{n+1} x^n + \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} 2^n x^n$$

$$\Rightarrow \frac{1}{x^2} \sum_{n=0}^{\infty} a_{n+2} x^{n+2} - \frac{2}{x} \sum_{n=0}^{\infty} a_{n+1} x^{n+1} + \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} 2^n x^n$$

$$\Rightarrow \frac{1}{x^2} [G(x) - a_0 - a_1 x] - \frac{2}{x} [G(x) - a_0] + G(x) = \sum_{n=0}^{\infty} 2^n x^n$$

$$\Rightarrow \frac{1}{x^2} [G(x) - 2 - x] - \frac{2}{x} [G(x) - 2] + G(x) = 1 + 2x + 2^2 x^2 + 2^3 x^3 + \dots$$

$$\Rightarrow G(x) \left[\frac{1}{x^2} - \frac{2}{x} + 1 \right] - \frac{(2+x)}{x^2} + \frac{4}{x} = (1-2x)^{-1}$$

$$\begin{aligned} \Rightarrow G(x) \left(\frac{1-2x+x^2}{x^2} \right) &= \frac{1}{1-2x} + \frac{2+x}{x^2} - \frac{4}{x} \\ &= \frac{1}{1-2x} + \frac{2+x-4x}{x^2} \\ &= \frac{1}{1-2x} + \frac{2-3x}{x^2} \end{aligned}$$

$$G(x) = \frac{x^2}{(1-x)^2} \left[\frac{1}{1-2x} + \frac{2-3x}{x^2} \right]$$

$$= \frac{x^2}{(1-2x)(1-x)^2} + \frac{2-3x}{(1-x)^2}$$

$$= \frac{x^2}{(1-2x)(1-x)^2} + \frac{3}{1-x} - \frac{1}{(1-x)^2}$$

$$G(x) = \frac{x^2}{(1-2x)(1-x)^2} + \frac{3}{1-x} - \frac{1}{(1-x)^2}$$

$$\swarrow$$

$$\frac{A}{1-2x} + \frac{B}{1-x} + \frac{C}{(1-x)^2}$$

$$A=1, B=0, C=-1$$

$$G(x) = \frac{1}{1-2x} - \frac{1}{(1-x)^2} + \frac{3}{1-x} - \frac{1}{(1-x)^2}$$

$$= (1-2x)^{-1} - 2(1-x)^{-2} + 3(1-x)^{-1}$$

$$= 1 + 2x + 2^2x^2 + \dots + 2^n x^n +$$

$$- 2[1 + 2x + 3x^2 + \dots + (n+1)x^n + \dots]$$

$$+ 3[1 + x + x^2 + \dots + x^n + \dots]$$

$$a_n = 2^n - 2(n+1) + 3$$

$$= 2^n - 2n + 1; \quad n \geq 0$$

Ex $a_n = 4a_{n-1} - 4a_{n-2} + 4^n, \quad n \geq 2$
 $a_0 = 2, \quad a_1 = 8$

Sol Let $G(x) = \sum_{n=0}^{\infty} a_n x^n$

$$a_n x^n - 4a_{n-1} x^n + 4a_{n-2} x^n = 4^n x^n$$

$$\sum_{n=2}^{\infty} a_n x^n - 4 \sum_{n=2}^{\infty} a_{n-1} x^n + 4 \sum_{n=2}^{\infty} a_{n-2} x^n = \sum_{n=2}^{\infty} 4^n x^n$$

$$\sum_{n=2}^{\infty} a_n x^n - 4x \sum_{n=2}^{\infty} a_{n-1} x^{n-1} + 4x^2 \sum_{n=2}^{\infty} a_{n-2} x^{n-2} = \sum_{n=2}^{\infty} 4^n x^n$$

$$(G(x) - a_0 - a_1 x) - 4x(G(x) - a_0) + 4x^2 G(x) = \frac{(4x)^2}{1-4x}$$

$$G(x) [1 - 4x + 4x^2] - a_0 - a_1 x + 4a_0 x = \frac{16x^2}{1-4x}$$

$$G(x) [1 - 4x + 4x^2] - 2 - 8x + 8x = \frac{16x^2}{1-4x}$$

$$G(x) [1 - 2x]^2 = \frac{16x^2}{1-4x} + 2 = \frac{16x^2 + 2 - 8x}{1-4x}$$

$$G(x) = \frac{(16x^2 - 8x + 2)}{(1-2x)^2 (1-4x)} = \frac{(1-4x)^2 + 1}{(1-2x)^2 (1-4x)}$$

$$\frac{A}{1-2x} + \frac{B}{(1-2x)^2} + \frac{C}{1-4x}$$

$$A = 0, B = -2, C = 4$$

$$G(x) = \frac{-2}{(1-2x)^2} + \frac{4}{1-4x} = -2(1-2x)^{-2} + 4(1-4x)^{-1}$$

$$= -2[1 + 2(2x) + 3(2x)^2 + \dots + (n+1)(2x)^n + \dots] + 4[1 + 4x + (4x)^2 + \dots + (4x)^n + \dots]$$

$$\sum_{n=0}^{\infty} a_n x^n$$

$$a_n = -2(n+1)2^n + 4 \cdot 4^n$$

$$\Rightarrow a_n = 4^{n+1} - (n+1)2^{n+1}, \quad n \geq 2$$